

High-Lift Airfoil Synthesis by Inverse Design and a Moving-Mesh Drag-Reduction Study

Ashray Saxena

Department of Mechanical Engineering, BITS Pilani, India

f20231043@pilani.bits-pilani.ac.in

Abstract

Short-field and blown-wing unmanned aircraft need high-lift devices that are both aerodynamically effective and mechanically deployable. This report covers two related pieces of aerodynamic work. The first, SkyNET, parameterizes airfoils with a Class Shape Transformation and solves an inverse-design problem for the coefficients that meet target lift and drag, evaluated with a viscous–inviscid panel method and the SU2 flow solver; the high-lift devices are designed together with their deployment kinematics, and the blown slot is shaped using the curvature ratio h/R so the jet stays attached. The second is a transient, arbitrary Lagrangian–Eulerian moving-mesh study of a drag-reduction system, which resolves how the slot flow and wake reorganize as the upper element opens and quantifies the resulting change in aerodynamic load.

Keywords: high-lift aerodynamics; inverse design; Class Shape Transformation; fowler flap; Coanda effect; moving mesh; ALE

I Introduction

Aircraft built for short-field and blown-wing operation need far more lift at low speed than a clean wing can supply, and the high-lift devices that deliver it must be both aerodynamically effective and mechanically deployable. The two are not independent: a flap that produces excellent lift but cannot be stowed, or whose slot separates as it deploys, is useless. This report covers two related pieces of aerodynamic work that together span the problem from shape to mechanism. The first, SkyNET, is an inverse-design and optimization framework for high-lift airfoils and their deployment paths. The second is a transient moving-mesh study of a drag-reduction system (DRS), an aerofoil whose upper element opens on demand to shed load.

II Airfoil parameterization and inverse design

The airfoils are represented with a Class Shape Transformation, in which the surface coordinate is the product of a class function and a Bernstein shape function,

$$y(x) = C(x)S(x), \quad C(x) = x^{N_1}(1-x)^{N_2}, \quad (1)$$

$$S(x) = \sum_{i=0}^n b_i \binom{n}{i} x^i (1-x)^{n-i}, \quad (2)$$

so that a compact set of coefficients b_i spans a rich family of shapes (Fig. 3). Rather than optimizing forward from a guessed shape, SkyNET works by *inverse design*: it solves for the coefficients that meet target lift and drag coefficients by minimizing

$$\mathcal{J} = (C_l - C_{l0})^2 + \lambda (C_d - C_{d0})^2, \quad (3)$$

with the aerodynamic coefficients evaluated by a coupled viscous–inviscid panel method in the XFOIL family [2] and, where compressibility matters, by the SU2 flow solver [3]. A parametric sweep over the design variables (Fig. 1) then selects the best-performing high-lift configuration.

III Deployable high-lift devices

A high-lift airfoil is only useful if it can be stowed and deployed, so the geometry of a single-slotted fowler flap with a Coanda vane is designed together with its kinematics (Fig. 2). The deployment path is built so that the lower surface remains free of inflection by construction (Fig. 4), and the slot is shaped to keep the blown jet attached. Attachment is governed by the local ratio of slot height to surface radius of curvature, h/R : the jet follows the curved surface only while

$$\frac{h}{R} \lesssim \left(\frac{h}{R}\right)_{\text{crit}}, \quad (4)$$

beyond which it separates. Tracking the radius of curvature and h/R along the Coanda arc (Figs. 5, 6) keeps the turning within that limit, and the deflected-jet states (Figs. 7, 8) show the resulting flow at increasing flap deflection. The nose contour is redesigned in tandem so the stowed configuration closes cleanly (Fig. 9).

IV Moving-mesh drag-reduction system

The DRS is a multi-element wing whose upper element rotates open to stall the wing and shed downforce. Modelling it faithfully requires the geometry to move during the solve, so the flow is computed in an arbitrary Lagrangian–Eulerian (ALE) frame. On a mesh deforming with velocity $\mathbf{u}_m$, a transported quantity ϕ evolves as

$$\frac{\partial \phi}{\partial t} \Big|_{\mathbf{x}} + (\mathbf{u} - \mathbf{u}_m) \cdot \nabla \phi = \mathcal{D}(\phi), \quad (5)$$

where $\mathbf{x}$ are the mesh reference coordinates and $\mathcal{D}$ collects the flux and source terms. A time-accurate solution on the deforming grid (Figs. 10, 11) resolves how the slot flow and the wake reorganize as the flap opens, and quantifies the change in aerodynamic load between the closed and open states.

V Conclusion

Together the two studies span the high-lift problem from shape to mechanism: an inverse-design framework that hits target lift and drag with deployable, attachment-aware high-lift devices, and a moving-mesh method that captures the transient aerodynamics of an actuated drag-reduction system. Both feed directly into the design of blown-wing and short-field unmanned aircraft.

USB Double-Slotted Fowler Flap – Parametric Sweep
NACA 64A415, 1000mm chord, F1=300mm, F2=350mm

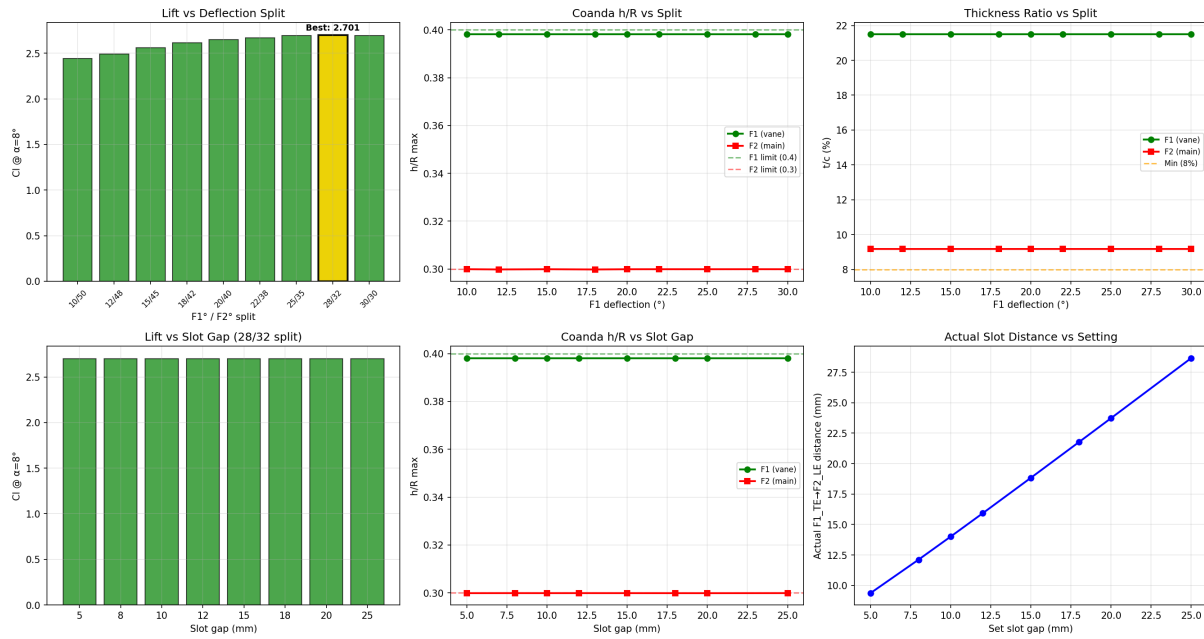


Figure 1. Parametric sweep of the high-lift configuration: lift against deflection and slot-gap settings.

References

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Fowler Flap v1.0.21 - SINGLE-SLOTTED, Coanda Vane Optimised
F1: Coanda vane (12mm, TE=1.5mm) delta = 50 deg c_{eff} = 1.31c

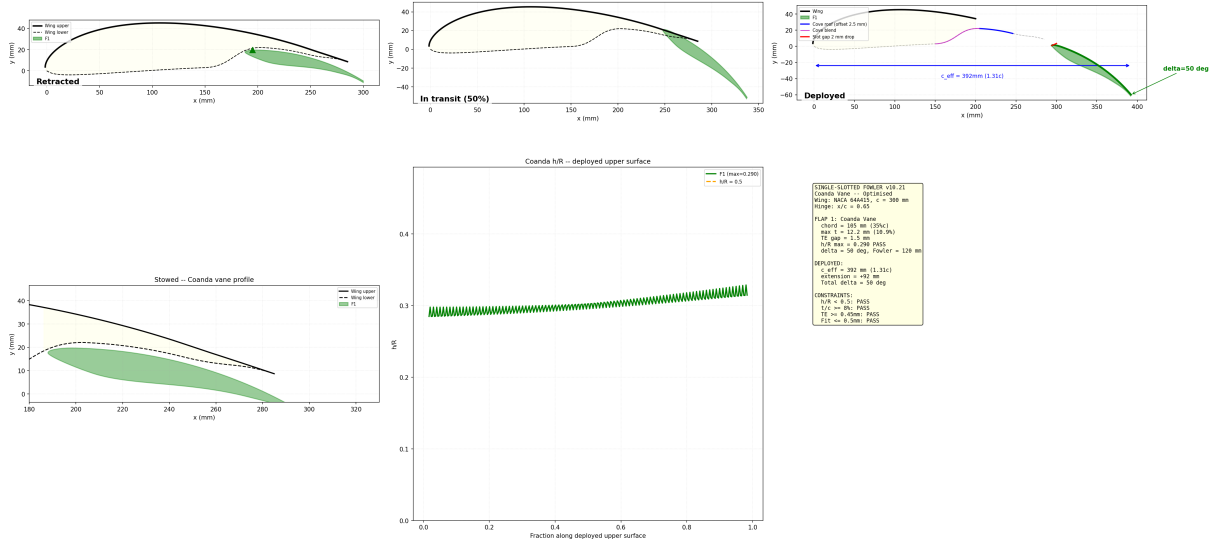
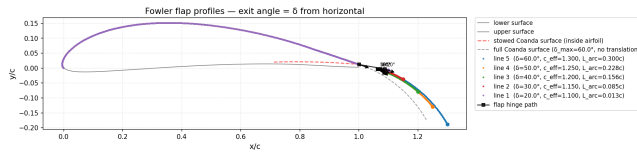


Figure 2. Single-slotted fowler flap with a Coanda vane, shown retracted, in transit, and fully deployed.



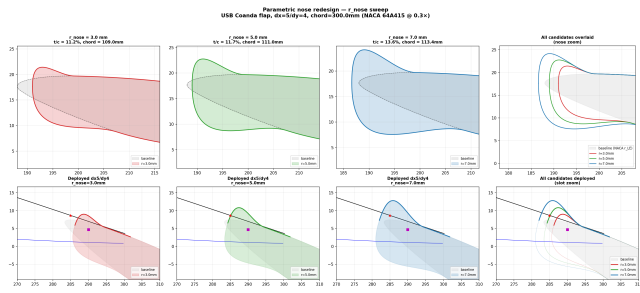


Figure 9. Nose redesign so the stowed configuration closes cleanly.

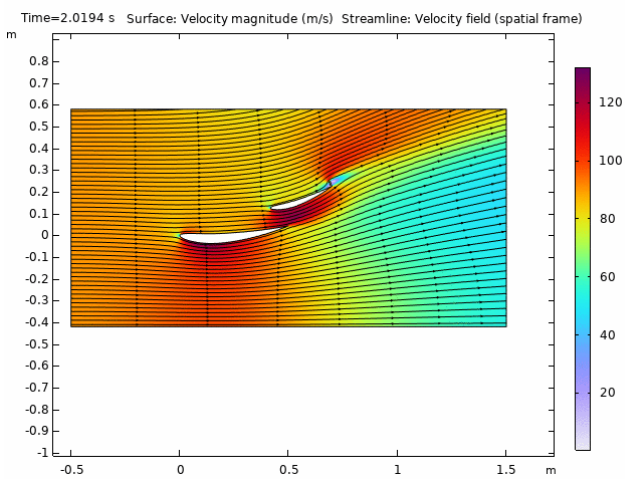


Figure 10. Velocity field and streamlines over the drag-reduction system as the upper element opens.

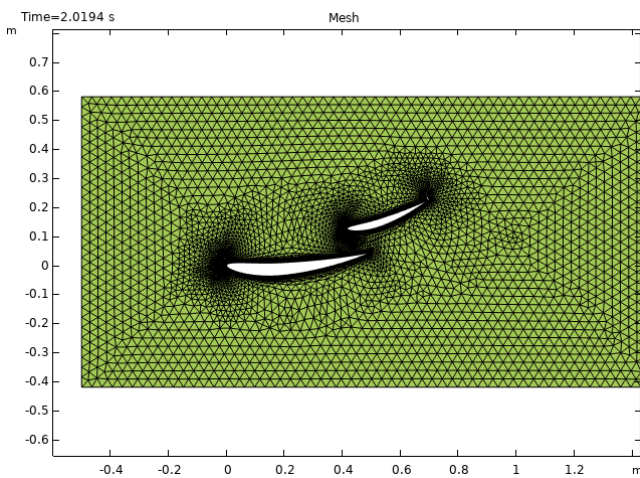


Figure 11. Deforming mesh tracking the two elements through actuation, refined where gradients are sharpest.